

Matteo Tusoni

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About me

PhD student in Computer Science at Sapienza University of Rome, working at the intersection of machine learning, causal discovery, time series analysis, and agent-based modelling. My research focuses on developing AI methods for understanding complex dynamical systems, with applications to financial and macroeconomic systems.

Education

PhD in Computer Science

Nov 2024–Nov 2027

Sapienza University of Rome

Rome, Italy

- **Description:** Research on machine learning, causal discovery, time series analysis, and agent-based modelling for financial and macroeconomic systems.
- **Advisor:** Prof. Novella Bartolini.

MSc in Theoretical Physics

Mar 2023

Sapienza University of Rome

Rome, Italy

- **Description:** Thesis on Higgs-boson decays into $\bar{b}b$ and $\bar{c}c$ with the ATLAS detector, including the development of machine learning methods for particle-flow reconstruction.
- **Advisor:** Prof. Marumi Kado; **Co-advisor:** PhD Francesco Armando di Bello.
- **Grade:** 110/110 cum laude.

Research Experience

Department of Computer Science, University of Oxford

Feb 2026 – May 2026

Visiting researcher

Oxford, UK

- **Research:** Developed and analyzed differentiable approaches to agent-based model simulations for economic and financial systems.
- **Team:** Research team led by Prof. Michael Wooldridge and Prof. Anisora Calinescu.

Applied Research Team, Bank of Italy

Jan 2025 – Aug 2025

PHD internship

Rome, Italy

- **Research:** Studied causal inference techniques for macroeconomic time series, focusing on structural dependencies in high-dimensional financial systems.
- **Research:** Explored multi-agent models for macroeconomic simulation and emergent behaviour in complex adaptive economies.

CERN – ATLAS Experiment

Mar 2022 – Mar 2023

Member, internship

Geneva, Switzerland

- **Research:** Contributed to AI-based particle detection, detector simulation, and studies of the Higgs boson within the ATLAS collaboration.
- **Contribution:** Contributed to ATLAS Note ANA-HIGG-2020-20-INT1 on Higgs-boson decays into $\bar{b}b$ and $\bar{c}c$ in VH production.

INFN – Frascati National Laboratories

Mar 2021 – Aug 2021

Member, internship

Frascati, Italy

- **Research:** Researched plasma acceleration and electron-beam interaction with ionized gas within the SPARC_LAB collaboration.

Publications

C=Conference/Workshop, J=Journal, U=Under review

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| [U] | Micro-Level Priors Enable Macro-Level Learning of Neural Agent-Based Models | 2026 |
| | <i>Under review – First author</i> | |
| [C] | Robust Causal Discovery in Real-World Time Series with Power-Laws | 2026 |
| | <i>ICML 2026 – Spotlight. Top 2% of papers – First author</i> | |
| [C] | Causal Discovery for Stochastic Power Law Time Series | 2025 |
| | <i>AIFIN 2025, ECAI Satellite Workshop – First author</i> | |
| [C] | Placy – Power law causal Discovery | 2025 |
| | <i>Complexity in Economics & Finance, StatPhys29 Satellite Workshop – First author</i> | |

- [J] **Reconstructing particles in jets using set transformer and hypergraph prediction networks** 2023
European Physical Journal C – Co-author
- [J] **Configurable calorimeter simulation for AI applications** 2023
Machine Learning: Science and Technology – Co-author
- [C] **HGPflow: Particle reconstruction as hyperedge prediction** 2022
Machine Learning and the Physical Sciences, NeurIPS 2022 Satellite Workshop – Co-author

Awards & Selected Programs

Machine Learning Summer School NYC 2026 Jun 2026
Columbia University New York, USA

- Fully funded participant in a competitive two-week residential programme in advanced machine learning.

G-Research – Spring into Quant Finance Apr 2026
G-Research London, UK

- Fully funded participant, selected among the top 1% of applicants for a programme in quantitative finance and research.

Sapienza's Call for Scientific Research Grant Jan 2026
Sapienza University of Rome Rome, Italy

- Competitive seed funding for an independent research project.

Professional Experience

Senior Data Scientist May 2024 – Nov 2024
Ernst & Young S.p.A. Rome, Italy

- **Data Science:** Delivered end-to-end data science projects for major clients, including Costa Crociere and Barilla, developing statistical and machine learning models and optimization methods for complex business problems.
- **Leadership:** Coordinated and managed the data science team within the group, contributing to the development of its technical capabilities and expertise.

Data Scientist & AI Specialist Jun 2023 – May 2024
Leonardo S.p.A. Rome, Italy

- **Secure & Explainable AI:** Developed a platform for secure and explainable AI, supporting the evaluation and interpretation of machine learning models.
- **Quantum Computing:** Designed and developed a traffic simulation using a quantum annealing algorithm to minimize traffic congestion.
- **GCAP:** Developed AI models for the Global Combat Air Programme (GCAP), a sixth-generation combat aircraft programme involving Italy, the United Kingdom, and Japan.
- **EDOCC:** Contributed to the definition and development of AI capabilities for the European Defence Operational Collaborative Cloud (EDOCC), an EU-funded federated cloud platform for multi-domain military operations.

Technical Skills

Programming	Python, C++, C, R, Julia, MATLAB, SQL
Machine Learning	Deep learning, PyTorch, TensorFlow, scikit-learn, Keras
Methods	Causal discovery, causal inference, time series analysis, agent-based modelling
Tools	Git, Docker, ROOT
Operating Systems	Linux, macOS, Windows
Languages	Italian (native), English (advanced), French (basic)